

# Frequency-domain training data for a learned plasma fluid closure

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## Abstract

We close the warm-beam moment hierarchy of a cold-bulk / warm-beam bump-on-tail system with a small MLP that replaces the Hammett–Perkins / Hunana Padé rational closure of  $Z'(\xi_b)$ . The architecture is standard; the contribution is the training-data design. The default approach — run many linearized kinetic simulations across a  $(u_b/v_b, \varepsilon, k)$  grid and harvest  $(\text{ratios}, \alpha)$  pairs from trajectories — gives a closure that is already a respectable improvement over Padé:  $\gamma_{\text{eff}}$  within  $\sim 10\%$  across most of the parameter grid, degrading to  $\sim 50\%$  at the marginal corner. Sampling  $\xi_b$  directly in a  $\mathbb{C}$  rectangle — the equivalent frequency-domain operation, evaluated in closed form via the Faddeeva function — gives the same NN  $10^{-4}$ – $10^{-3}$  relative error nearly everywhere, two decades tighter than the sim-trained closure and three to four decades tighter than Padé. The frequency-domain dataset is also  $\sim 100\times$  cheaper to generate (one Faddeeva eval per sample, no kinetic time-stepping, no IC choices) and is problem-agnostic in  $(u_b/v_b, \varepsilon, k)$ . The geometric reason the win exists: linear eigenmodes from a kinetic simulation trace a narrow region in the  $\xi_b$ -plane, leaving the rest of the relevant rectangle uncovered.

## 1 Setup

A cold background plasma (density  $n_0$ ) coupled to a warm Maxwellian beam (density  $n_b = \varepsilon n_0$ , drift  $u_b$ , thermal speed  $v_b$ ). In beam-frame coordinates ( $s = (v_x - u_b)/v_b$ ) the linearized two-fluid system has state  $y = (u, E, U_0, U_1, U_2)$  with

$$\partial_t u = -E, \tag{1}$$

$$\partial_t E = u - \varepsilon(u_b U_0 + U_1), \tag{2}$$

$$\partial_t U_n = -iku_b U_n - ikU_{n+1} + nM_{n-1}E, \quad n < 2, \tag{3}$$

where  $U_n = (1/n_b) \int s^n f_{b1} ds$  and  $M_k = \langle s^k \rangle_{f_{b0}}$  ( $M_0 = 1$ ,  $M_1 = 0$ ,  $M_2 = 1/2$ ,  $\dots$ ). The closure problem is to specify  $U_3$  at each step.

The kinetic linear response of the beam to a potential  $\hat{\Phi}$  is

$$U_0/\hat{\Phi} = Z'(\xi_b), \quad \xi_b = (\omega - ku_b)/(kv_b), \tag{4}$$

with  $Z(\xi) = i\sqrt{\pi} w(\xi)$  the plasma dispersion function and  $w$  the Faddeeva function. Iterating the recurrence  $\xi_b U_n - U_{n+1} = nM_{n-1}\hat{\Phi}$  from (4) gives the exact 3-moment closure ratio

$$\alpha(\xi_b) \equiv U_3/U_0 = \xi_b^3 - \xi_b/Z'(\xi_b). \tag{5}$$

The kinetic dispersion at scales much longer than the Debye length is

$$D_{\text{kin}}(\omega, k) = 1 - 1/\omega^2 - (\varepsilon/k^2)Z'(\xi_b) = 0. \tag{6}$$

**Padé closures.** A linear-in-moments closure  $U_3 = \sum_{i<3} a_i U_i$  with constants  $a_i$  combined with the moment recurrence yields a degree-(1, 3) rational approximant of  $Z'(\xi_b)$ . Choosing  $(a_i)$  to match the Taylor series of  $-\sqrt{\pi}Z'$  at  $\xi_b = 0$  to order  $\xi_b^N$  is the Padé closure  $R_N$  in the Hammett–Perkins / Hunana hierarchy.

## 2 Padé baseline

Figure 1 compares the closed fluid response of the Padé hierarchy  $N \in \{3, 4, 5, 6\}$  to the kinetic  $-\sqrt{\pi}Z'(\xi_b)$  along real  $\xi_b$ . Each Padé closure matches the kinetic curve to  $\sim 10^{-2N+3}$  at  $\xi_b = 0$  (Taylor matching) and degrades rapidly to residual of order unity for  $|\xi_b| \gtrsim 1$ . The imaginary part (Landau resonance) is captured at only  $\sim 30\%$  of its kinetic amplitude even at  $N = 6$ . This is the fundamental Padé ceiling: a finite-order rational approximates an entire function only locally.

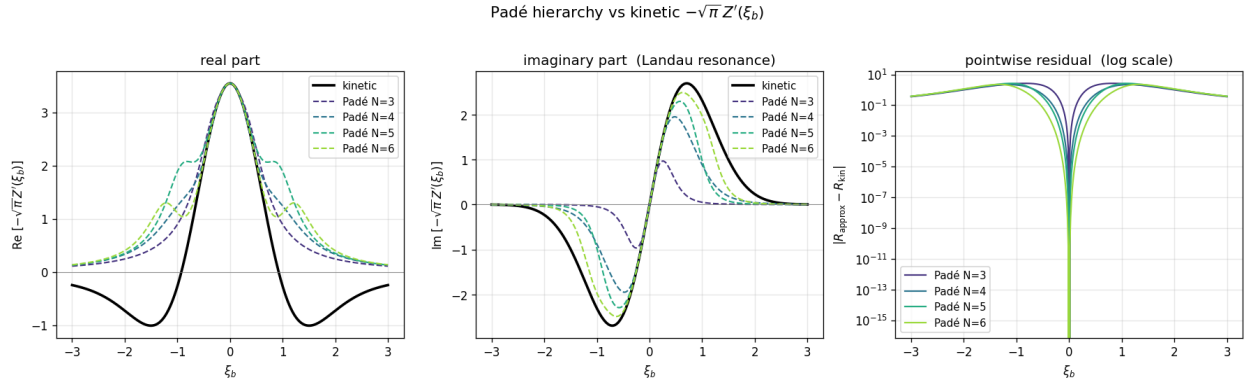


Figure 1: Padé hierarchy vs kinetic  $-\sqrt{\pi}Z'(\xi_b)$  along real  $\xi_b$ . Real part (left), imaginary part / Landau resonance (middle), pointwise residual on log scale (right). Padé residuals plateau at  $\sim 10^0$  for  $|\xi_b| > 1$ .

Figure 2 plots the relative error in the BoT  $\gamma_{\max}$  at the canonical operating point ( $u_b/v_b = 5, \varepsilon = 0.05$ ) vs Padé order. The hierarchy oscillates around the kinetic answer, crossing zero between  $N = 7$  and  $N = 8$ .

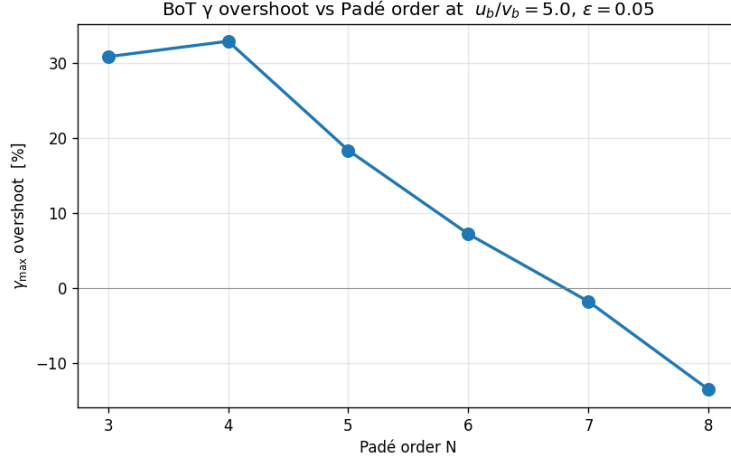


Figure 2:  $\gamma_{\max}$  overshoot at  $(u_b/v_b, \varepsilon) = (5, 0.05)$  vs Padé order. Non-monotone — crosses zero near  $N = 7$ .

### 3 A closed-form identity for the per-mode closure

Before turning to the NN we record one closed-form identity. It fixes what *any* closure must reduce to on a single linear eigenmode, and will let us state precisely what the NN does and does not learn (Section 9).

Applying the moment recurrence  $U_{n+1} = \xi_b U_n - n M_{n-1} \hat{\Phi}$  for  $n = 0, 1, 2$  with  $M_0 = 1$ ,  $M_1 = 0$ :

$$\begin{aligned} U_1 &= \xi_b U_0, \\ U_2 &= \xi_b U_1 - M_0 \hat{\Phi} = \xi_b^2 U_0 - \hat{\Phi} \quad (\text{since } M_0 = 1), \\ U_3 &= \xi_b U_2 - 2M_1 \hat{\Phi} = \xi_b U_2 \quad (\text{since } M_1 = 0). \end{aligned}$$

The third line eliminates the inhomogeneous coupling because the first velocity moment of a Maxwellian vanishes in its own frame. Dividing by  $U_0$  and using  $\xi_b = U_1/U_0$ :

$$\boxed{\alpha = \frac{U_3}{U_0} = \frac{U_1}{U_0} \cdot \frac{U_2}{U_0}} \quad (7)$$

i.e. on any linear eigenmode the closure ratio is the complex product of the two moment ratios. Direct check:  $r_1 r_2 = \xi_b (\xi_b^2 - 1/Z'(\xi_b)) = \xi_b^3 - \xi_b/Z'(\xi_b) = \alpha(\xi_b)$ . All the plasma-dispersion machinery is absorbed into  $U_2$ ; the closure step itself is multiplication.

**Used as a closure, this fails.** On a superposition  $U_n(t) = \sum_i A_i e^{-i\omega_i t} U_n^{(i)}$  the identity breaks: the product of sums in  $r_1 \cdot r_2$  generates cross-terms ( $i \neq j$ ) that the per-mode formula  $\sum_i A_i \xi_{bi} U_2^{(i)}$  does not. So hardcoding  $U_3 = U_1 U_2 / U_0$  in a fluid solver is not a usable closure. At the canonical  $(5, 0.05, k = 0.24)$  point, with a Langmuir-projected IC (single eigenmode), bilinear matches the kinetic  $\gamma_{\text{eff}}$  to six decimals; with a generic  $\delta E$  IC (all five fluid eigenmodes) it gives +28.4% overshoot — essentially Padé-level (+30.8%).

The same picture holds across the full sweep (Figure 3, middle panel): bilinear is a strict improvement on Padé in every cell, but the improvement is small (8–14% relative in the regimes where Padé is acceptable; larger only at the marginal corner where both are deep into failure).

In practical terms it is not a useful alternative to Padé. Padé and the Hunana hierarchy chose linear-in-moments closures precisely so that they commute with superposition; (7) breaks that property and inherits the cost.

**Why we record it anyway.** The identity (7) fixes what the per-mode closure equals, in closed form. The NN closure ( $f_\theta : (r_1, r_2) \mapsto \alpha$ ) is structurally free to be anything; the identity gives a baseline against which  $f_\theta$  can be compared on and off the eigenmode manifold. Section 9 uses it to show that the NN reproduces the bilinear product on the manifold but a distinctly non-multiplicative function off it, and that the off-manifold behaviour is what makes the closure usable.

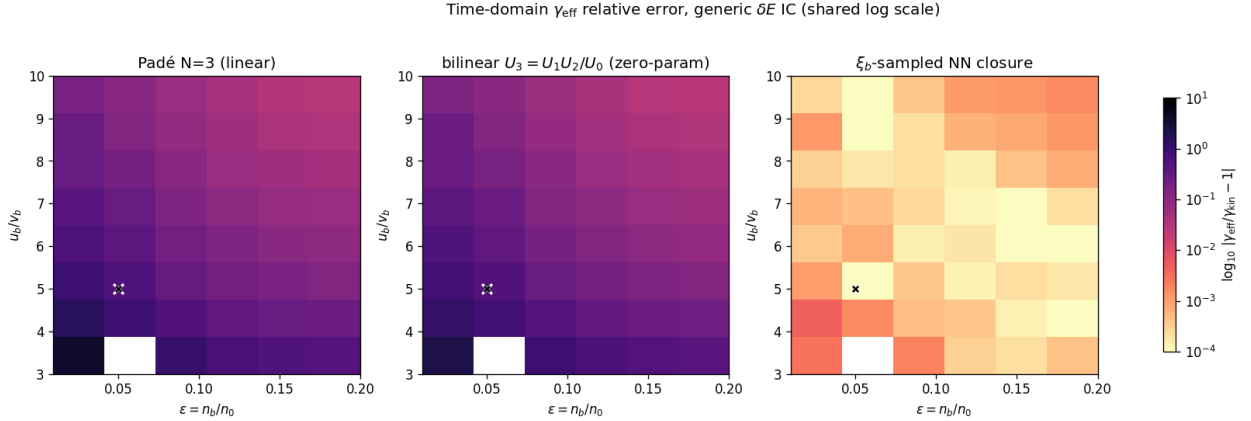


Figure 3: Time-domain  $\gamma_{\text{eff}}$  relative error across the  $(u_b/v_b, \varepsilon)$  grid from a generic  $\delta E$  IC, on a shared log scale. *Left*: Padé  $N = 3$ . *Middle*: bilinear identity  $U_3 = U_1 U_2 / U_0$  used as a closure (Section 3). *Right*:  $\xi_b$ -sampled NN closure (Section 8). The bilinear identity is a strict but small improvement on Padé in every cell — both are in the  $10^{-1}$ – $10^0$  regime. The NN sits two to three decades tighter.

## 4 NN closure

We replace the Padé linear closure with a small MLP  $f_\theta : (U_1/U_0, U_2/U_0) \rightarrow \alpha$  that outputs the closure ratio directly:

$$U_3 = f_\theta(U_1/U_0, U_2/U_0) \cdot U_0. \quad (8)$$

This is a nonlinear-in-moments closure with no rational-order ceiling: NN expressivity bounds the approximation error, not the polynomial degree. The architecture is fixed throughout: 4-layer MLP, (64, 64, 64, 64) hidden, tanh activations,  $\sim 8.7\text{k}$  parameters. Input: 4 reals (re/im of two complex moment ratios). Output: 2 reals (re/im of  $\alpha$ ). Adam, cosine LR  $3 \times 10^{-3} \rightarrow 10^{-5}$ , 8000 steps, batch 512. What changes between the experiments below is the *training data*.

## 5 First attempt: simulation-trained closure

The default training-data choice is the same one used everywhere else in data-driven closures: integrate the kinetic system, harvest moments  $(U_0, U_1, U_2, U_3)$  along trajectories, regress  $U_3/U_0$  on  $(U_1/U_0, U_2/U_0)$ . We build a broad sweep: 24 operating points  $(u_b/v_b \in \{3, 4, 5, 6, 8, 10\}) \times (\varepsilon \in \{0.02, 0.05, 0.10, 0.20\})$ , five  $k$  values per operating point, 12 random initial conditions each. Total:  $\sim 10^5$  training samples after filtering transient outliers. The sweep is intentionally broad to

maximise coverage. This is a substantial amount of effort:  $24 \times 5 = 120$  kinetic integrations, with mode-projection bookkeeping at each timestep.

**Linear dispersion.** Figure 4 compares the sim-trained NN to the kinetic  $\gamma(k)$  at four representative  $(u_b/v_b, \varepsilon)$  points. The closure tracks the kinetic dispersion curve through the unstable branch at three of the four; peak- $\gamma$  overshoot is below 1% near the centre of the training set and grows to +12% at the marginal corner  $(3, 0.05)$ . The spikes in damped- $k$  regions are artefacts of fitting  $\gamma$  from a generic-IC trajectory where no unstable mode exists — the late-time signal is dominated by whichever fluid eigenmode is least damped, and the sim-trained closure produces inaccurate spurious modes there.

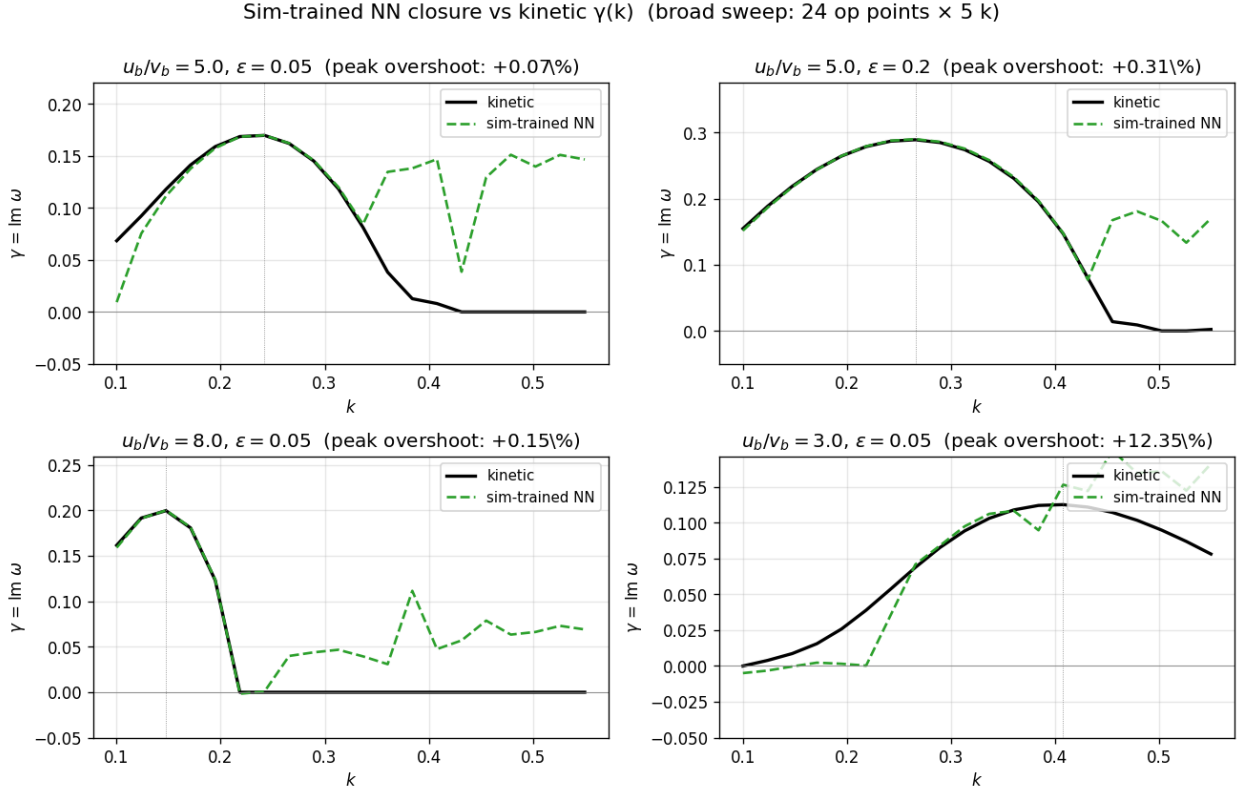


Figure 4:  $\gamma(k)$  for the simulation-trained NN closure (green dashed) vs kinetic (black) at four representative operating points. The sweep spans the training distribution; dispersion match is  $\lesssim 1\%$ .

**Time-domain at canonical.** Figure 5 integrates the sim-trained closure at the canonical operating point under both a Langmuir-projected and a generic- $\delta E$  IC. The closure tracks the kinetic envelope to graphical precision in both cases; fitted  $\gamma_{\text{eff}} = +0.1701$  vs  $\gamma_{\text{kin}} = +0.1700$ .

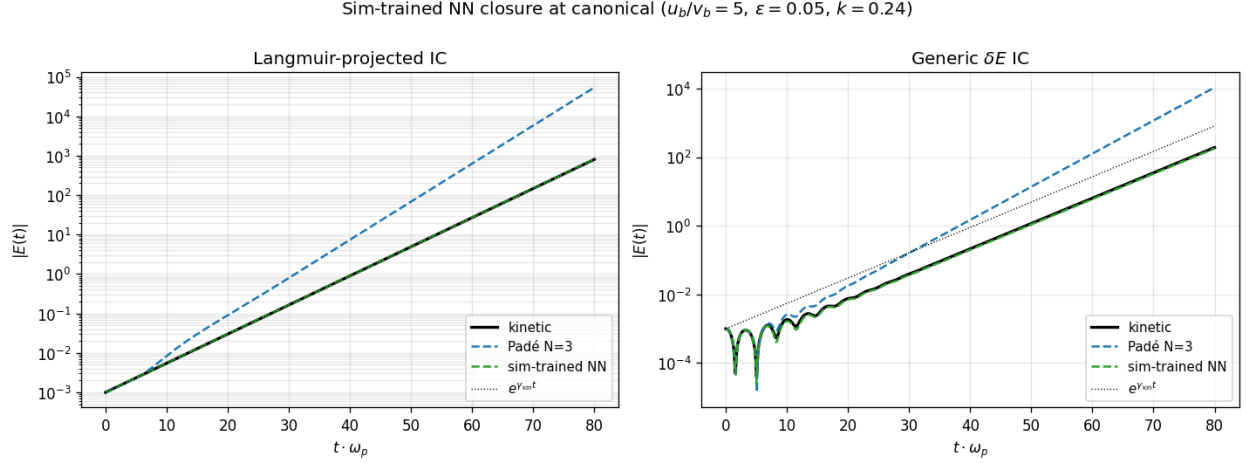


Figure 5: Sim-trained NN closure at canonical ( $u_b/v_b = 5$ ,  $\varepsilon = 0.05$ ,  $k = 0.24$ ). Tracks the kinetic envelope under both ICs — this operating point is well within the training distribution.

**2D parameter scan.** Figure 6 shows the time-domain  $\gamma_{\text{eff}}$  relative error from a generic  $\delta E$  IC across the  $(u_b/v_b, \varepsilon)$  grid. The colour scale is  $\log_{10} |\gamma_{\text{eff}}/\gamma_{\text{kin}} - 1|$  from  $10^{-4}$  to  $10^1$ , shared with Figures 8 and 10. Padé sits at  $10^{-1}$ – $10^0$ . The sim-trained NN is roughly one decade better than Padé in the well-trained interior of the grid and stays at  $10^{-2}$ – $10^{-1}$  over most cells, with the marginal corner reaching  $\sim 10^{-0.3}$ . This is genuinely a serviceable closure — within  $\sim 10\%$  on  $\gamma_{\text{eff}}$  across most of the operating regime, which is what one would hope to get from a data-driven approach trained on  $10^5$  simulation samples.

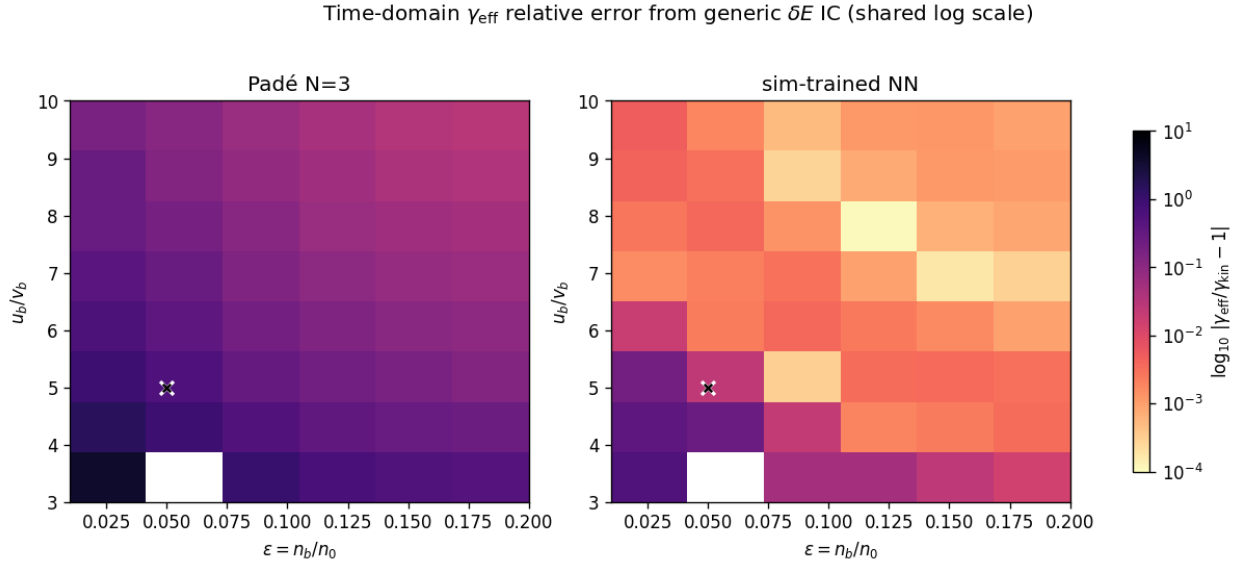


Figure 6: Time-domain  $\gamma_{\text{eff}}$  relative error for Padé (left) and the simulation-trained NN closure (right) on a shared log scale. The sim-trained closure is a decade better than Padé in the well-trained interior, with the largest residuals at the low- $\varepsilon$  low- $u_b$  corner.

The natural question is whether one can do better without simply running more simulations.

## 6 Diagnosis: 1D coverage in $\xi_b$

The closure problem is fundamentally one-parameter: on a linear eigenmode,  $\alpha$  is a function of  $\xi_b$  alone (5), and  $(U_1/U_0, U_2/U_0)$  are too. So we can plot any training dataset as a distribution in the  $\xi_b$ -plane.

Figure 7 does this. The orange rectangle is the ranges of  $\xi_b$  relevant for unstable BoT modes ( $\text{Re } \xi_b \in [-3, 3], \text{Im } \xi_b \in [-1, 1.5]$ ). The green scatter is the  $\xi_b$  value on the most-unstable kinetic mode as  $(u_b/v_b, \varepsilon, k)$  is varied across a fine grid covering and exceeding the simulation training set. The scatter occupies a narrow sub-region of the rectangle: it is bounded to  $\text{Im } \xi_b > 0$  (unstable modes have positive Landau-resonance imaginary part) and is heavily concentrated near a 1-parameter curve in the upper-left quadrant.

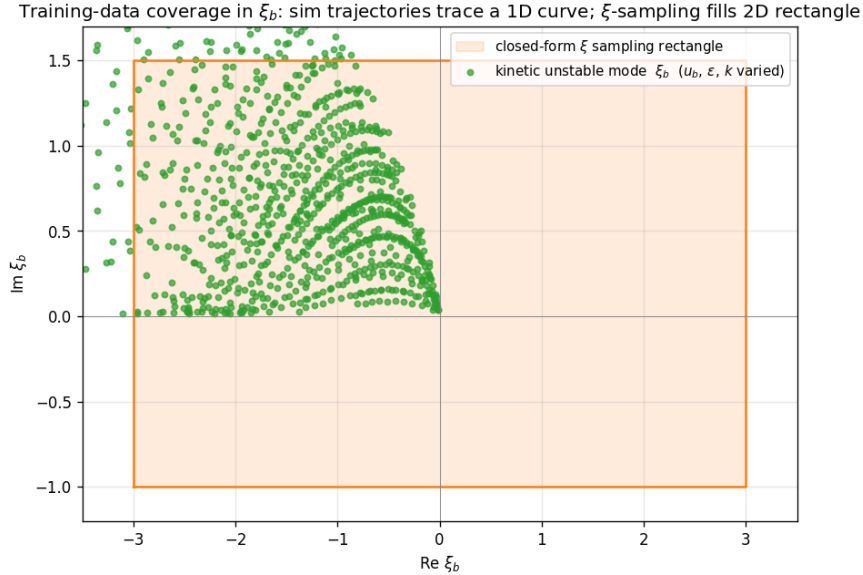


Figure 7:  $\xi_b$  on the most-unstable kinetic mode as  $(u_b/v_b, \varepsilon, k)$  is varied (green); closed-form  $\xi_b$ -sampling rectangle (orange). The orange rectangle covers all relevant  $\xi_b$  including damped/spurious modes; the green region covers only the unstable branch.

**Why this matters.** The 5-d closed fluid system has *five* eigenmodes, only one of which is the Langmuir-like unstable mode. A generic- $\delta E$  IC excites a superposition of all five. Their projected  $\xi_b$  values are spread across the rectangle — many of them outside the green region. When the NN sees moment ratios from those modes it has no training data nearby and outputs an extrapolated  $\alpha$ . If the extrapolated  $\alpha$  happens to overstabilise the spurious mode the trajectory still settles onto Langmuir-like growth (this is why canonical single-trace tests look fine); if it doesn't — as in the marginal corner — the spurious modes contaminate  $\gamma_{\text{eff}}$ .

This is structural. No amount of additional simulation training data fills the rectangle: kinetic trajectories live on the eigenmode manifold, and the eigenmode manifold's projection in  $\xi_b$ -space *is* the green region. Adding more  $(u_b/v_b, \varepsilon, k)$  adds more samples on the same set.

## 7 Fix: $\xi_b$ -sampling

The kinetic linear theory provides an analytical handle. On a linear eigenmode the forward map  $\xi_b \mapsto (U_1/U_0, U_2/U_0)$  is closed-form: starting from  $U_0/\hat{\Phi} = Z'(\xi_b)$  and iterating the moment recurrence,

$$U_0 = Z'(\xi_b), \quad U_1 = \xi_b Z'(\xi_b), \quad U_2 = \xi_b^2 Z'(\xi_b) - 1, \quad (9)$$

$$U_1/U_0 = \xi_b, \quad U_2/U_0 = \xi_b^2 - 1/Z'(\xi_b). \quad (10)$$

Combined with (5), each tuple  $((\text{Re}, \text{Im } U_1/U_0, \text{Re}, \text{Im } U_2/U_0), (\text{Re}, \text{Im } \alpha))$  is one Faddeeva evaluation. *Same equations as the simulation training; same single underlying kinetic linear theory.* The only difference is that we sample  $\xi_b$  directly (frequency-domain) rather than reconstructing it from time-domain trajectories.

We replace the simulation dataset with 16,384  $\xi_b$  values uniformly distributed in  $\text{Re } \xi_b \in [-3, 3]$ ,  $\text{Im } \xi_b \in [-1, 1.5]$  — the orange rectangle in Figure 7. The training and architecture are otherwise identical.

**Data-generation cost.** The simulation dataset of §4 required 120 kinetic integrations across the operating sweep ( $\sim 10^5$  filtered samples; minutes of wall time on a single core including the mode-projection bookkeeping). The closed-form dataset is 16,384 `scipy.special.wofz` calls — a fraction of a second of wall time, no operating-point sweep, no ICs, no transient outliers to filter. The ratio is roughly two orders of magnitude in compute and a much larger ratio in design effort.

**$\alpha$  accuracy at test points.** Table 1 compares  $\alpha$  prediction error at representative  $\xi_b$  values for the sim-trained (single op pt), sim-trained (broad, 24 op pts) and closed-form-trained closures. The first column is well inside all training distributions and all three closures work. The remaining rows are off the unstable-mode curve — moment ratios from damped or spurious modes. Sim training fails by 50–800%; closed-form training is uniformly sub-1%.

| $\xi_b$                               | sim (1 op pt) | sim (broad, 24 op pts) | closed-form |
|---------------------------------------|---------------|------------------------|-------------|
| $-1.17 + 0.71i$ (canonical eigenmode) | +2.3%         | +0.8%                  | +0.05%      |
| $-0.83 + 0.05i$                       | +148%         | +62%                   | +0.5%       |
| $0.50 + 0.00i$                        | +843%         | +247%                  | +0.6%       |
| $-1.50 - 0.20i$                       | +50%          | +27%                   | +0.2%       |
| $1.20 + 0.10i$                        | +209%         | +118%                  | +0.6%       |
| $-2.00 + 0.00i$                       | +13%          | +7%                    | +0.5%       |

Table 1:  $|\alpha_{\text{NN}} - \alpha_{\text{exact}}|/|\alpha_{\text{exact}}|$  at representative  $\xi_b$ . Sim sampling fits well only at the unstable Langmuir mode; closed-form is accurate everywhere.

## 8 Results with $\xi_b$ -sampling

**Linear dispersion sweep.** Figure 8 compares the  $\gamma_{\text{max}}$  relative error  $|\gamma_{\text{fluid}} - \gamma_{\text{kin}}|/\gamma_{\text{kin}}$  across the  $(u_b/v_b, \varepsilon)$  grid on the same shared log scale as the previous section. Padé  $N=3$  sits at  $10^{-1}$ – $10^0$  (worst at marginal, matching Figure 2). The  $\xi_b$ -sampled NN closure is  $\leq 10^{-3}$  across nearly the entire grid — two to four decades tighter than the sim-trained closure (Figure 6) and three to four decades tighter than Padé.

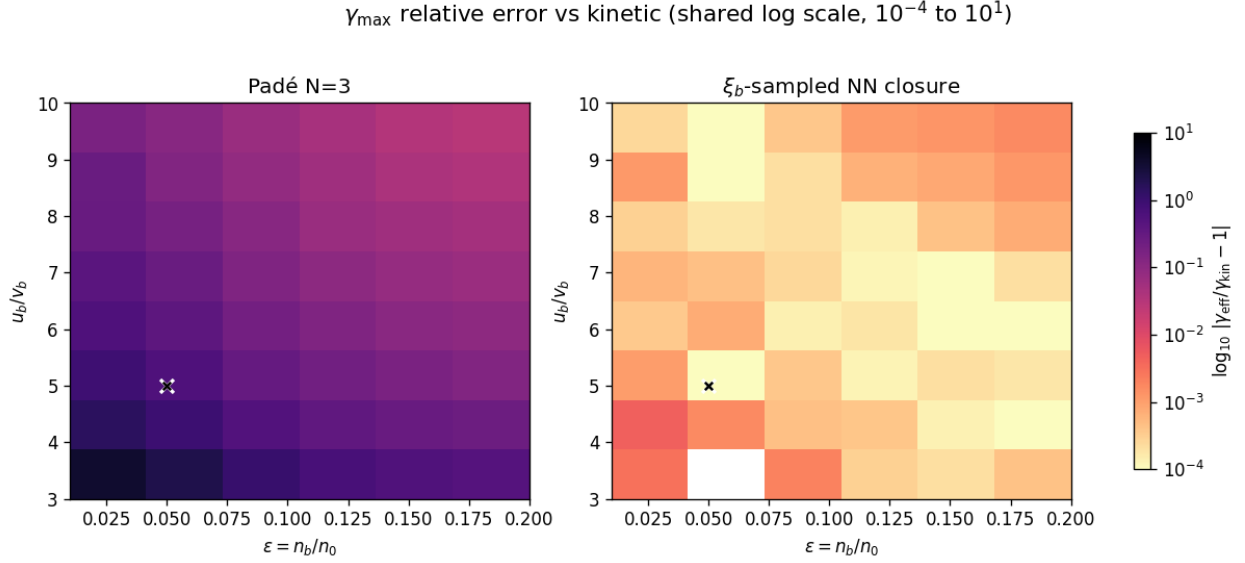


Figure 8:  $\gamma_{\max}$  relative error vs kinetic across the  $(u_b/v_b, \varepsilon)$  grid. Padé  $N=3$  (left) vs  $\xi_b$ -sampled NN closure (right) on the shared log scale. White  $\times$  marks canonical  $(5, 0.05)$ .

Table 2 reports four representative points.

| $(u_b/v_b, \varepsilon)$ | $\gamma_{\text{kin}}$ | Padé $N = 3$ | $\xi_b$ -sampled NN |
|--------------------------|-----------------------|--------------|---------------------|
| canonical $(5, 0.05)$    | 0.170                 | +30.8%       | −0.02%              |
| hot beam $(5, 0.20)$     | 0.289                 | +13.9%       | +0.02%              |
| strong drift $(8, 0.05)$ | 0.200                 | +11.2%       | +0.01%              |
| marginal $(3, 0.05)$     | 0.113                 | +107.9%      | +0.20%              |

Table 2:  $\gamma_{\max}$  overshoot at representative operating points.  $\sim 10^3$ – $10^4 \times$  improvement in relative error vs Padé  $N = 3$ .

**Time-domain canonical stress test.** Figure 9 integrates all three systems for  $t \in [0, 80/\omega_p]$  at the canonical operating point, with Langmuir-projected and generic- $\delta E$  ICs. Fitted  $\gamma_{\text{eff}}$ :  $\gamma_{\text{kin}} = +0.1700$ ,  $\gamma_{\text{pade}} = +0.2223$ ,  $\gamma_{\text{NN}} = +0.1700$ .

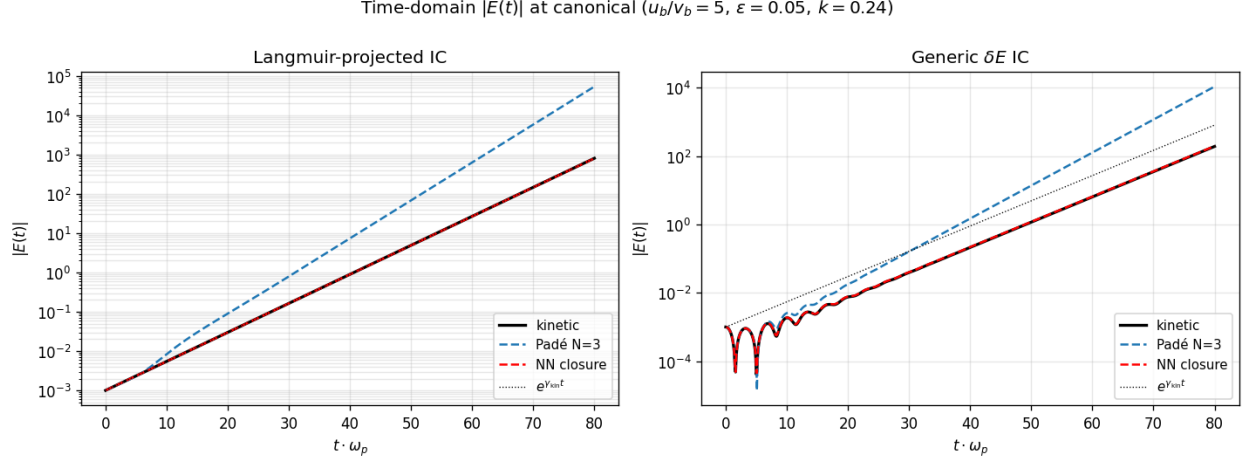


Figure 9: Time-domain  $|E(t)|$  at canonical ( $5, 0.05, k = 0.24$ ) under Langmuir-projected and generic  $\delta E$  initial conditions.  $\xi_b$ -sampled NN closure (red dashed) tracks kinetic (black) in both cases.

**Time-domain across parameter space.** Figure 10 repeats the experiment for each  $(u_b/v_b, \varepsilon)$  cell. The  $\xi_b$ -sampled closure is  $\lesssim 0.1\%$  over most of the grid; the marginal corner shows mild under-prediction (NN training rectangle's lower-left isn't quite reached in  $\xi_b$ ). No blow-up anywhere. The contrast with Figure 6 is the headline: the same NN architecture, trained on the same physics but sampled in frequency rather than time, fixes the marginal-corner failure.

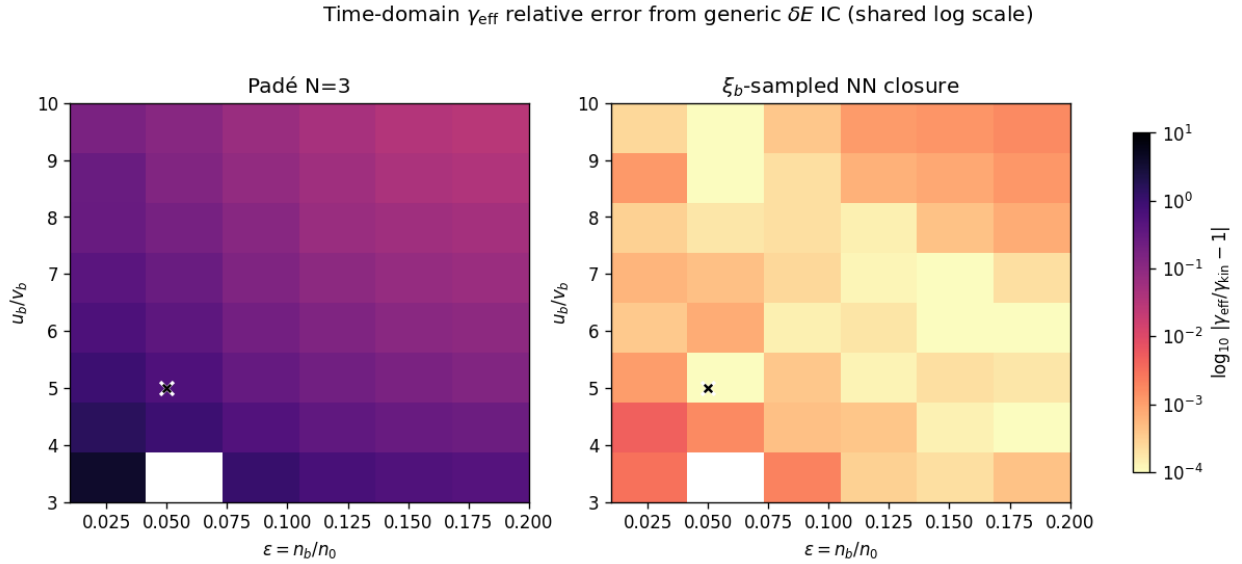


Figure 10: Time-domain  $\gamma_{\text{eff}}$  relative error from generic  $\delta E$  IC across the parameter grid: Padé (left) vs  $\xi_b$ -sampled NN closure (right) on the same shared log scale as Figure 6. The contrast against Figure 6 is the headline: same NN architecture, same fluid integrator, same generic IC, but now  $10^{-4}$ – $10^{-3}$  across nearly the whole grid.

## 9 What did the NN actually learn?

Section 3 gives the structurally correct closure on a single eigenmode in closed form:  $\alpha = r_1 \cdot r_2$ . The NN was trained on  $\xi_b$ -sampled data, i.e. exclusively on on-manifold points. A natural question is whether the NN simply learned multiplication. The answer has two parts: yes on the manifold, no off it; and the off-manifold extrapolation is what makes the closure work.

**On the eigenmode manifold.** Sampling  $\xi_b$  uniformly in the training rectangle and evaluating  $f_\theta(r_1(\xi_b), r_2(\xi_b))$  against  $r_1 r_2$  gives a mean relative error of  $\sim 3 \times 10^{-3}$ , consistent with the NN’s training loss. So on the 2-real-dimensional submanifold the NN reproduces the bilinear product within training precision.

**Off the manifold.** Sampling  $r_1$  and  $r_2$  *independently* in the bounding box of the on-manifold cloud — points that do not correspond to any single  $\xi_b$  — gives a mean relative error of order one between  $f_\theta(r_1, r_2)$  and  $r_1 r_2$ . The NN does not learn complex multiplication globally; it learned multiplication on the manifold and a smooth (but distinctly non-multiplicative) interpolation off it.

**Why this matters: bilinear vs NN in time-domain.** Figure 11 shows the consequence. At two operating points (canonical and marginal), we integrate four fluid systems – kinetic ground truth, NN closure, bilinear closure, Padé — from a generic  $\delta E$  IC. Top row:  $|E(t)|$ . The bilinear trajectory tracks Padé, not kinetic.

The bottom row isolates the closure function from any trajectory feedback: along the kinetic trajectory, we evaluate  $\alpha_{\text{NN}}(r_1^{\text{kin}}, r_2^{\text{kin}})$  and  $\alpha_{\text{bil}} = r_1^{\text{kin}} r_2^{\text{kin}}$  and compare both to the kinetic ground truth  $U_3^{\text{kin}}/U_0^{\text{kin}}$ . Two phases appear:

- *Early time* ( $t \lesssim 20\omega_p^{-1}$  at canonical;  $t \lesssim 50$  at marginal): the fluid state is a real superposition of eigenmodes,  $(r_1, r_2)$  sit off the manifold, and the bilinear product misses  $\alpha_{\text{kin}}$  by  $10^{-1}$ – $10^0$ . The NN sits at  $\sim 10^{-2}$  throughout.
- *Late time*: the unstable mode dominates,  $(r_1, r_2)$  relax onto the manifold, and the bilinear product becomes exact ( $\sim 10^{-5}$  relative error). The NN plateaus at its  $\sim 10^{-2}$  training residual.

*Bilinear is the better closure function in the asymptotic single-mode limit; the NN is the better closure function in the early multimode phase.* Because closure errors at early time seed spurious fluid eigenmodes that grow exponentially, early-time accuracy dominates the final  $\gamma_{\text{eff}}$ . The NN’s off-manifold smoothness is what locks the trajectory onto the kinetic eigenmode before spurious-mode contamination matters; hardcoded multiplication is the algebraic extension off the manifold and is the wrong extension.

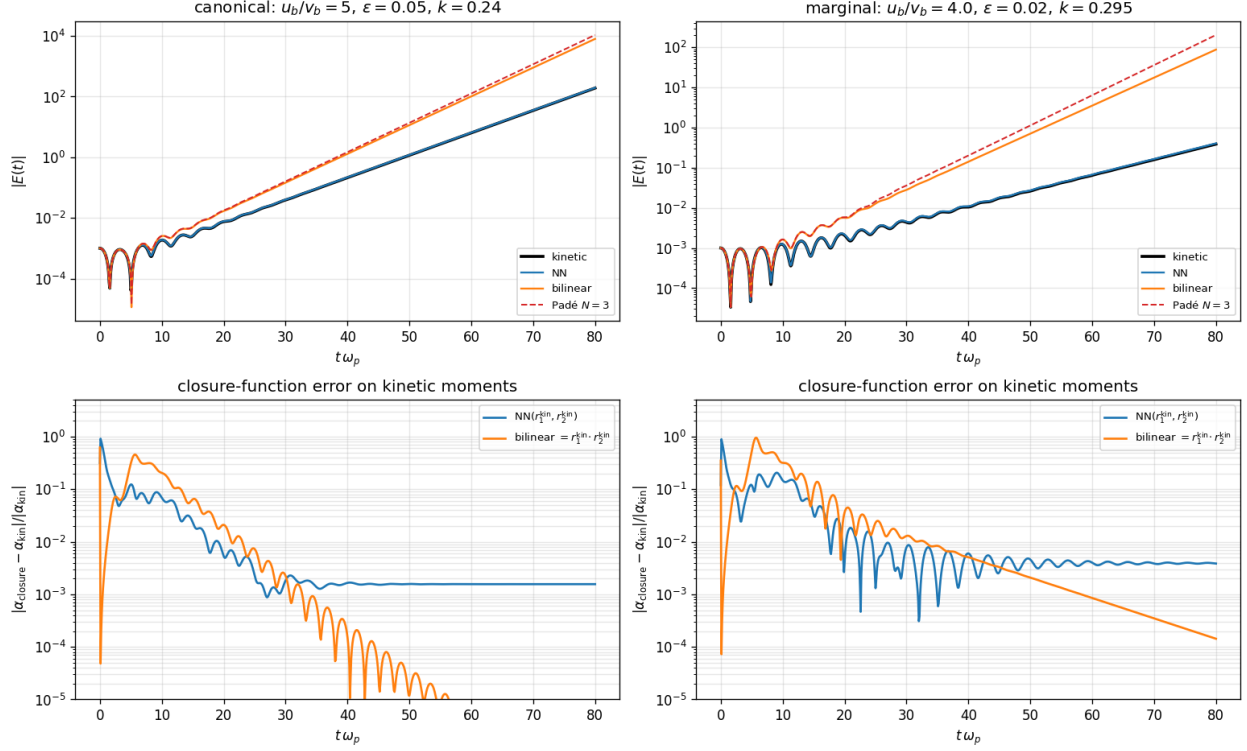


Figure 11: Off-manifold diagnostic at canonical (left) and marginal (right) operating points. *Top*:  $|E(t)|$  for kinetic / NN / bilinear / Padé under a generic  $\delta E$  IC. Bilinear (orange) visibly tracks Padé, not kinetic. *Bottom*: closure-function error  $|\alpha_{\text{closure}} - \alpha_{\text{kin}}|/|\alpha_{\text{kin}}|$  evaluated on the kinetic moments, isolating the closure function from trajectory feedback. NN: blue. Bilinear: orange. Early time ( $t \lesssim 20$  canonical,  $\lesssim 50$  marginal) the NN is one to two decades better than bilinear; once the unstable mode dominates, bilinear is exact and the NN plateaus at training residual.

**Mechanism, in one line.** On the eigenmode manifold the 3-moment Maxwellian closure is bilinear and exact. The NN reproduces that on the manifold and supplies a useful smooth extension off it — useful in the precise sense that it suppresses superposition cross-term error at early time and lets the trajectory relax onto the manifold. Linear closures (Padé/HP) miss both the on-manifold and the off-manifold structure; bilinear gets the on-manifold structure but fails on the off-manifold extension; the NN gets both.

## 10 Generalization: Landau damping on a single warm Maxwellian

The trained closure  $f_\theta(r_1, r_2) \rightarrow \alpha$  depends only on the Maxwellian linear response: nothing in the training set references  $u_b$ ,  $\varepsilon$ , or  $k$ . So the same checkpoint should work for any 3-moment fluid system on a Maxwellian equilibrium, not just BoT. This section confirms it by computing Landau damping of Langmuir waves on a single warm Maxwellian — a textbook benchmark unrelated to the BoT instability.

**Setup.** Strip the BoT system to a single warm species (drop the cold bulk equations, set  $\varepsilon = 1$ ,  $u_b = 0$ ):

$$\begin{aligned}\partial_t E &= -U_1, \\ \partial_t U_0 &= -ikU_1, \\ \partial_t U_1 &= -ikU_2 + E \quad (M_0 = 1), \\ \partial_t U_2 &= -ikU_3 \quad (M_1 = 0, \text{ closure}).\end{aligned}$$

The kinetic Langmuir dispersion in this normalization is  $k^2 = Z'(\omega/k)$ , related to the standard  $k\lambda_D$  convention by  $k_{\text{code}} = \sqrt{2} k\lambda_D$  (BoT-paper variance is  $1/2$ , so  $v_{th} = 1/\sqrt{2}$ ).

**Method.** For each  $k\lambda_D$ :

1. Solve  $k^2 = Z'(\omega/k)$  for the kinetic Langmuir mode (Newton from the Bohm-Gross seed) – analytic ground truth.
2. Project initial state onto the kinetic Langmuir eigenvector:  $(E, U_0, U_1, U_2) = (U_1/i\omega, 1, \xi, \xi^2 - 1/Z'(\xi))$  with  $\xi = \omega_{\text{kin}}/k$ .
3. Integrate the NN-closed and Padé-closed fluid systems in time, fit  $(\omega_r, \gamma)$  from late-time log-magnitude slope and real-part zero crossings.

Projecting onto the kinetic eigenvector isolates the Langmuir mode – without it, the fluid system’s spurious closure eigenmodes (which exist because there is no cold-bulk grounding) dominate the late-time signal.

**Result.** Figure 12 shows  $(\omega_r, |\gamma|)(k\lambda_D)$  on  $k\lambda_D \in [0.30, 0.60]$ .

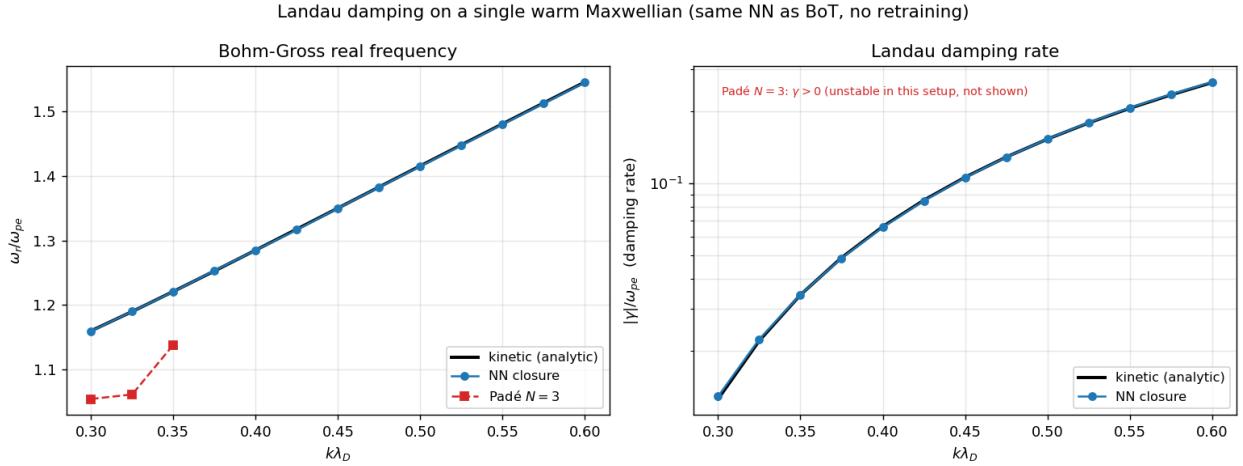


Figure 12: Landau damping of Langmuir waves on a single warm Maxwellian, using the same NN checkpoint trained for BoT. *Left*: Bohm-Gross real frequency. *Right*: damping rate, log scale. The NN closure reproduces the kinetic ground truth (textbook values:  $\gamma/\omega_{pe} = -0.0126$  at  $k\lambda_D = 0.3$ ,  $-0.1534$  at  $k\lambda_D = 0.5$ , etc.) to better than 1% across the range. Padé  $N = 3$  in this setup gives the wrong frequency and is unstable ( $\gamma > 0$ ); not shown on the log-damping axis. See text.

The NN curve overlays the kinetic curve in both panels. Representative values:

| $k\lambda_D$ | kinetic $\gamma/\omega_{pe}$ | NN $\gamma/\omega_{pe}$ | rel. error |
|--------------|------------------------------|-------------------------|------------|
| 0.30         | -0.0126                      | -0.0129                 | 2.4%       |
| 0.40         | -0.0661                      | -0.0657                 | 0.6%       |
| 0.50         | -0.1534                      | -0.1535                 | 0.07%      |
| 0.60         | -0.2641                      | -0.2655                 | 0.5%       |

**Why Padé fails here.** The same Hunana-style Padé closure that gave  $\sim 30\%$  overshoot on BoT  $\gamma_{\text{eff}}$  produces an *unstable* fluid system on the pure warm Maxwellian:  $\gamma > 0$  even when initialized on the kinetic Langmuir eigenvector. This is a known consequence of Taylor-matching closures in systems without cold-fluid grounding; Hammett-Perkins-style frequency-axis filtering can fix it but is a separate construction. The NN closure has no such problem because the per-mode bilinear identity it learned (Section 3) gives the exact Langmuir eigenmode by construction: the closed fluid system’s Langmuir eigenvalue equation reduces to  $Z'(\omega/k) = k^2$  identically.

**What this means.** The closure object stored in `naive_mlp.eqx` is a learned Maxwellian linear-response function. It is reusable for any 3-moment fluid integrator with a Maxwellian equilibrium and  $\xi_b$  in the trained rectangle. BoT is one instance, Landau damping on a warm Maxwellian is another, and the same logic extends to ion-acoustic waves with adiabatic electrons, drift waves with adiabatic species, or any model where the closure is ”evaluate  $Z'$  of an effective  $\xi_b$ ”. A separate trained model is needed only when the equilibrium is non-Maxwellian (kappa, flat-top, trapped plateau) or when going beyond 3 moments (the per-mode identity at higher order is not just a product because  $M_2 \neq 0$ ).

## 11 Discussion

**Where the win comes from.** Both wins — accuracy and cost — trace to the same observation: the closure problem is parameterised by a single complex variable  $\xi_b$ . Simulation training generates (ratios,  $\alpha$ ) pairs via the parameterisation  $(u_b/v_b, \varepsilon, k, \text{IC}, t) \mapsto \xi_b$ , which is many-to-few and covers only a 1-parameter family of  $\xi_b$  values (the unstable eigenmode), at the cost of a kinetic integration per operating point. Frequency-domain training generates the same pairs via the identity  $\xi_b \mapsto (\text{ratios}, \alpha)$ , which is one-to-one, covers whatever subset of  $\mathbb{C}$  we choose, and reduces dataset generation to a Faddeeva evaluation. The 2D rectangle is the natural, dense, problem-agnostic training distribution because moment ratios from the 5-d fluid system’s other (damped) eigenmodes also land in this rectangle.

The sim-trained closure is not bad —  $\sim 10\%$  on  $\gamma_{\text{eff}}$  across most of the grid is a real improvement over Padé. The honest framing is that the closed-form approach is strictly better on every axis that matters: tighter error (two decades), cheaper data (two orders of magnitude), problem-agnostic (no operating-point sweep), and trivially re-trained when the operating regime shifts.

The same logic applies to other moment closures whose linear-response function is one-parameter (drift-kinetic with one resonance, weakly collisional fluid models on a Maxwellian background): if there is an analytical  $\xi_b \mapsto \alpha$  map, frequency-domain sampling is strictly better than simulation sampling at fixed dataset size.

**What is not solved.** The training is entirely linear-physics:  $\alpha(\xi_b)$  is derived assuming the beam perturbation is linear about the Maxwellian  $f_{b0}$ . Capturing trapping, plateau formation, and saturated BoT requires kinetic training data and is outside this scope. The  $\xi_b$ -rectangle covers the

unstable BoT regime by construction; the marginal corner ( $u_b/v_b \lesssim 3, \varepsilon \lesssim 0.02$ ) pushes  $\xi_b$  to the rectangle’s lower edge and shows mild degradation.

**Relation to the Padé / Hunana hierarchy.** Hunana et al. (2019) construct rational closures of  $Z'(\zeta)$  up to 8 poles and prove asymptotic convergence with increasing order. The NN closure trades the rational-order ceiling for NN expressivity; the present work shows the fluid response problem itself isn’t the obstacle — training-data design is.

## Code

The accompanying `bot/` package implements all figures and runs:

- `closures/pade.py` — Padé closure coefficients and the closed fluid response.
- `closures/naive_nn.py` — NN closure ( $\text{MLP} \rightarrow \alpha$ ), closed-form  $\xi_b$ -sampling dataset, training.
- `train_simdata.py` — sim-data NN training (single- and broad-sweep).
- `kinetic.py`, `fluid.py` — kinetic ground truth and two-fluid eigenvalue system.
- `closed_loop.py` — time-domain integration for all closures.
- `sweep.py`, `sweep_timedomain.py` — 2D parameter scans.
- `figures.py` — all plots in this writeup.